Rudolf Reiter, Dr.-Ing.

Research Scientist — Mechatronics, Optimization, Learning & Control

ORCID: 0009-0007-7635-2132 · Google Scholar: 5VdYugYAAAAJ

Zürich, Switzerland · rreiter@ifi.uzh.ch · rudolfreiter.github.io

Research and engineering at the intersection of optimal control, reinforcement learning, and autonomy. Experienced in leading multi-partner projects, proposal writing, and delivering robust software for real-world robotic systems (drones, autonomous vehicles, measurement systems).

Professional Experience

Postdoctoral Researcher — **Feb 2025 – Present**
University of Zürich, Robotics and Perception Group (PI: Prof. Davide Scaramuzza) — *Zürich, Switzerland*

- Research on agile vision-based autonomous drone navigation (learning/optimization/control).
- Acting lab lead deputy during PI sabbatical: oversaw day-to-day operations, staffing, and coordination.
- Project/funding leadership: authored, coauthored, and coordinated multi-institution proposals (Sony Research; Marie Skłodowska-Curie networks; Swiss infrastructure proposal; Zurich–Singapore proposal).
- Managed University of Zurich's part of EU flagship project AUTOASSESS (20+ industry/academic partners across Europe): leader board, WP lead, milestones, reporting
- Managed two industrial sub-projects (each €100k) originating from AUTOASSESS
- Led lab processes and external presence: organized/chaired lab meetings, led drone racing group, hosted visiting professors/industry, led outreach (Swiss Robotics Day, Zürich AI Summit), managed hiring team.

Postdoctoral Researcher — **Nov 2024 – Apr 2025**
University of Freiburg, Systems Control and Optimization Lab (PI: Prof. Moritz Diehl) — *Freiburg, Germany*

- Research on combining model predictive control with reinforcement learning (RL-guided MPC, pioneering survey and classification).
- Scientific supervision, project coordination across academic and industrial collaborators.

Doctoral Researcher (PhD) — **Mar 2020 – Nov 2024**
Systems Control and Optimization Laboratory, University of Freiburg (Advisor: Prof. Moritz Diehl) — *Freiburg, Germany*

- Dissertation: *Optimization-Based Motion Planning for Autonomous Driving and Racing*.
- Developed MPC and mixed-integer optimization methods for safe, high-performance motion planning in real-world settings.
- Supervision, project execution, and organization of research activities and large events.

Software Developer (Python/C++) (concurrent to PhD) — **Dec 2019 – Aug 2023**
Autonomous Racing Graz — *Graz, Austria*

- Embedded algorithms for real-world autonomous racing, emphasizing prediction, planning, and control.

Software Developer / Junior Researcher (Python/C++) — **Dec 2018 – Jul 2021**
Virtual Vehicle Research Center — *Graz, Austria*

- Path planning and control algorithms for autonomous vehicles; development of a simulation framework for autonomy stacks.

Control Systems Specialist (MATLAB/C/C++/Python) — **Jul 2016 – Jul 2018**
Anton Paar GmbH — *Graz, Austria*

- Lead development of advanced control systems for high-end measurement devices; delivered the first full automation of an atmospheric distillation analyzer; advanced nitrogen/convection heating of rheometer; developed an internal state estimator for microwave synthesizers; automated controller design for thermal analyzer.
- Inventor on US patent: *US 10,976,230 B2* (novel viscosity measurement).

Technical Skills

Core areas	Robotics, optimal control, machine learning, reinforcement learning, nonlinear / mixed-integer optimization, motion planning, estimation, autonomous systems
Programming	Python (7y), C/C++ (5y), MATLAB (6y)
Optimization / solvers	CasADi, acados, FORCES Pro, YALMIP, Gurobi, CVX, MOSEK
ML / robotics	PyTorch, JAX, Stable Baselines3, RLlib, OpenCV, Hydra; ROS; NVIDIA platforms; Linux; Speedgoat
Leadership	Team leadership and supervision, proposal writing, multi-partner project management, budgeting, hiring team lead, outreach/event leadership and organization
Languages	German (native), English (business fluent)

Education

Master of Science — Electrical Eng. (Control Systems & Mechatronics) **Jan 2013 – Jan 2016**
TU Graz / University of Utah · *GPA: 1.2/1.0 (with distinction)* *Graz, Austria / Salt Lake City, USA*

Bachelor of Science — Electrical Eng. (Control Systems & Mechatronics) **Oct 2009 – Oct 2012**
TU Graz · *GPA: 1.5/1.0 (with distinction)* *Graz, Austria*

Research Stays and Internships

Guest Researcher — ETH Zürich, Switzerland (Prof. Melanie Zeilinger)	4 months	Jan 2024
Guest Researcher — IMT Lucca, Italy (Prof. Alberto Bemporad)	1 month	Sep 2023
Research Intern — Mitsubishi Electric Research Laboratories (MERL), USA	4 months	Jan 2023
Research Intern — ODYS S.r.l., Italy	2 months	Apr 2022
Research Intern — Virtual Vehicle Research Center, Austria	8 months	Apr 2015
Controls Engineering Intern — B&R GmbH, Austria	5 months	Sep 2012

Talks and Organization of Scientific Events

Tutorial on Model Predictive Control & Reinforcement Learning @ ECC	>100 guests	Jul 2026
Keynote Tutorial @ ICRA	>8000 guests	Jun 2026
Workshop Co-Organizer: Aerial inspection for marine infrastructures @ ICRA	>100 guests	Jun 2026
Workshop Oral: AI-Driven Safe Aerial Robotics @ ICRA	>100 guests	Jun 2026
Organizer Booth @ Swiss Robotics Day Booth	>1000 guests	Nov 2025
Organizer Booth @ AI+X Summmit Zurich	>1000 guests	Oct 2025
Workshop Organizer on Ethical Innovation for Autonomous Systems	>50 guests	Oct 2024
Organizer of Advisory Board Meeting Marie Skłodowska-Curie Program	>20 guests	Oct 2022
Organizer of Workshop on Embedded Optimization and Learning for Robotics	>150 guests	Oct 2022
Master's Course on Model Predictive Control and Reinforcement Learning	>100 guests	Oct 2022
PhD Course on Model Predictive Control Theory and Computation	>100 guests	Jul 2022

Awards, Honors and Media

Academic Excellence Fellowship, University of Zurich	Award	Mar 2026
Magna Cum Laude — PhD Thesis, University of Freiburg	Honor	Nov 2024
Marie Skłodowska-Curie PhD Fellowship, European Commission	Award	Aug 2021
2nd place — ROBORACE Season Beta 1, Las Vegas Motor Speedway	Award	Mar 2021
2nd place — ROBORACE Season Beta 2, Las Vegas Motor Speedway	Award	Feb 2021
"Selbstfahrende Sieger" — Kleine Zeitung	Media	Sep 2020
Outstanding Success (Top 5%) — Master, TU Graz	Honor	Jan 2016
Talent Scholarship — AVL List GmbH	Award	Jan 2013
Outstanding Success (Top 2%) — Bachelor, TU Graz	Honor	Jul 2013
Performance Scholarship — TU Graz	Award	Jul 2011